

Jeel Chatrola

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EDUCATION

Worcester Polytechnic Institute

Master of Science - Robotics Engineering (GPA: 4.0/4.0)

Worcester, MA

Aug 2023 - May 2025

SKILLS SUMMARY

Programming Languages	Python, C++, C, MATLAB
Frameworks	PyTorch, TensorFlow, Scikit-learn, OpenCV, Open3D, PCL, MLflow, ROS/ROS2, MoveIt, OMPL, Ray
Tools	Bash/Zsh, Git, Linux, Docker, Bazel, Nix, Simulators (Gazebo, Pybullet, Isaac Sim, MuJoCo)
Hardware Platforms	Embedded Systems (Teensy, ESP32, Raspberry Pi, Nvidia Jetson), Stereo Camera, LIDAR, IMU, Turtlebot3, UR10
Other Skills	Version Control, Debugging and Profiling, Software Development, Functional Programming, Software Design Patterns, Documentation

WORK EXPERIENCE

Torc Robotics

Software Engineer I - Simulation

Ann Arbor, MI

Aug 2025 - Present

- Developing a shared **C/C++ simulation engine** with Python bindings and **Ray** support, powering both closed-loop scenario execution and reinforcement-learning rollouts at up to **200K steps per second**.
- Implemented performance optimizations that increased native scenario-engine throughput by **40%** for autonomous-driving validation workloads.
- Leading integration of the simulation stack across downstream Perception, Planning, and Controls use cases, including **CI-based validation workflows**.

Magna Services of America - Advance Manufacturing Innovation

Robotics Software Engineer Intern

Troy, MI

Aug 2024 - Jul 2025

- Developed an automated synthetic data generation pipeline using **Isaac Sim** and CAD models for training robust perception models for automotive components, generating a dataset with over **500M+ grasp labels**.
- Implemented and training a **PointNet++** based grasp detection network on synthetic point cloud data, achieving an estimated **80-85%** grasp success rate in simulation, improving upon baseline geometric methods.
- Designed the perception and grasping modules as reusable **ROS2** components with standardized interfaces for deployment on varied hardware platforms.
- Benchmarked model inference performance on target hardware (**NVIDIA Jetson**) to meet real-time processing requirements for industrial automation tasks.

Robert Bosch Centre for Cyber-Physical Systems, IISc Bangalore

Robotics Research Associate

Bangalore, India

Jan 2022 - Jun 2023

- Developed a comprehensive autonomous robotics framework in ROS2, integrating a real-time perception pipeline using **YOLOv7** and **DeepLabv3+** for object detection and semantic segmentation, optimized for **Jetson TX2** with 15FPS throughput.
- Implemented a multi-sensor perception stack integrating low-cost **LIDAR** and **stereo cameras** for ground robot navigation, utilizing **Kalman Filters** for sensor fusion to improve localization reliability by ~10% in dynamic environments.
- Managed a team of **20 interns** on research initiatives, establishing standardized documentation and testing procedures that improved project completion rates by 25% while ensuring long-term maintainability of deployed systems.
- Created comprehensive documentation and testing procedures for robotics software, ensuring maintainability and reliability of deployed systems.

PROJECTS

3D Scene Reconstruction: NeRF vs SfM 📄

Jun 2024 - Jul 2024

- Implemented and benchmarked Neural Radiance Fields (NeRF) against traditional Structure from Motion (SfM) for 3D reconstruction, achieving 12% better novel view synthesis with NeRF while SfM required 60% less computation time for comparable spatial accuracy (Python, OpenCV, COLMAP)

Automatic Camera Calibration

May 2024 - Jun 2024

- Implemented Zhang's Camera Calibration method for intrinsic and extrinsic parameter estimation and lens distortion correction, achieving sub-pixel re-projection error on calibration datasets
- Utilized OpenCV to process checkerboard images and optimize camera parameters, enhancing accuracy of downstream vision tasks (Python, OpenCV)

Panorama Stitching

May 2024 - Jun 2024

- Implemented advanced algorithms such as SIFT and SURF for feature extraction, brute force matching for feature alignment, and homography for image warping to combine multiple images into one (Python, OpenCV)

Vision based Obstacle avoidance for quad-rotors

Apr 2024 - May 2024

- Created a Deep-Reinforcement Learning based framework to perform end-to-end motion planning for quad-rotor using motion primitives (Python, PyTorch, OpenCV)

Mask R-CNN for Pose Estimation in Dynamic Environments

Nov 2023 - Dec 2023

- Fine-tuned a **Mask R-CNN** model with a **ResNet-50** backbone using **TensorFlow** on the **COCO dataset** for accurate 2D human keypoint detection.
- Developed a real-time feedback application using **OpenCV** to visualize pose estimation results, assisting users in maintaining correct exercise form with processing latency under 100ms. (Python, TensorFlow, Scikit-learn, OpenCV)

Path Planning Algorithms

Sep 2023 - Nov 2023

- Implemented 8+ path-finding algorithms from scratch, including BFS, DFS, Dijkstra's, PRM, RRT, RRT, and A* (Python, C++, OMPL)

Research Grade Modular Mobile Robot Platform

Jan 2023 - Jun 2023

- Developed a ROS-based hardware and software stack for a modular mobile robot, enabling algorithm development for perception and planning
- Implemented a closed-loop real-time controller in Embedded C with encoder feedback, achieving sub-10ms latency for precise speed and position control (ROS, Embedded C, Python)

Autonomous Navigation for Ground Robots using Symbolic Controllers

Jul 2022 - Dec 2022

- Engineered a safety-focused autonomy framework for mobile robots, incorporating a symbolic motion planner with formal guarantees, improving navigation efficiency by 35% compared to baseline RRT*
- Validated the framework in ROS/Gazebo simulations and on a Turtlebot3 platform, demonstrating robust performance in cluttered environments (ROS, Python, C++)

Multi-Robot Safe Planning using Control-Barrier Functions

May 2022 - Jul 2022

- Implemented Control Barrier Function (CBFs) layer to ensure 100% safety compliance for testing of diverse swarm/multi-agent algorithms with motion capture, enhancing experimentation and reliability (ROS, Python)

Vision Based Package Sorting using a UR5 Robotic Manipulator

Dec 2020 - Feb 2021

- Developed a ROS-based vision system for package sorting using a UR5 robot arm in a Gazebo simulated warehouse
- Implemented object detection using YOLOv5 for package identification. Integrated motion planning using MoveIt for efficient sorting tasks (ROS, Python, C++, OpenCV, MoveIt, Gazebo)

PUBLICATIONS

- **Multi-layer Motion Planning with Kinodynamic and Spatio-Temporal Constraints** 
ACM Hybrid System Control Conference (HSCC - 2025)
- **Autonomous Exploration using Ground Robots with Safety Guarantees** 
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS - 2023)
- **Scalable Controller for Multi-Agent Systems Using Barrier Functions and Symbolic Control** 
IEEE Conference on Decision and Control (CDC - 2023)
- **Funnel-based Reachability Control of Unknown Nonlinear System using Gaussian Process** 
IEEE Indian Control Conference (ICC - 2022)